

Chapter 1

Introduction

In this chapter we briefly discuss the role of modulation in a typical digital communication system, basic modulation methods, and criteria for choosing modulation schemes. Also included is a brief description of various communication channels, which will serve as a background for the later discussion of the modulation schemes.

1.1 DIGITAL COMMUNICATION SYSTEMS

Figure 1.1 is the block diagram of a typical digital communication system. The message to be sent may be from an analog source (e.g., voice) or from a digital source (e.g., computer data). The analog-to-digital (A/D) converter samples and quantizes the analog signal and represents the samples in digital form (bit 1 or 0). The source encoder accepts the digital signal and encodes it into a shorter digital signal. This is called source encoding, which reduces the redundancy hence the transmission speed. This in turn reduces the bandwidth requirement of the system. The channel encoder accepts the output digital signal of the source encoder and encodes it into a longer digital signal. Redundancy is deliberately added into the coded digital signal so that some of the errors caused by the noise or interference during transmission through the channel can be corrected at the receiver. Most often the transmission is in a high-frequency passband, the modulator thus impresses the encoded digital symbols onto a carrier. Sometimes the transmission is in baseband, the modulator is a baseband modulator, also called formator, which formats the encoded digital symbols into a waveform suitable for transmission. Usually there is a power amplifier following the modulator. For high-frequency transmission, modulation and demodulation are usually performed in the intermediate frequency (IF). If this is the case, a frequency up-converter is inserted between the modulator and the power amplifier. If the IF is too low compared with the carrier frequency, several stages of carrier frequency conversions are needed. For wireless systems an antenna is the final stage of the trans-

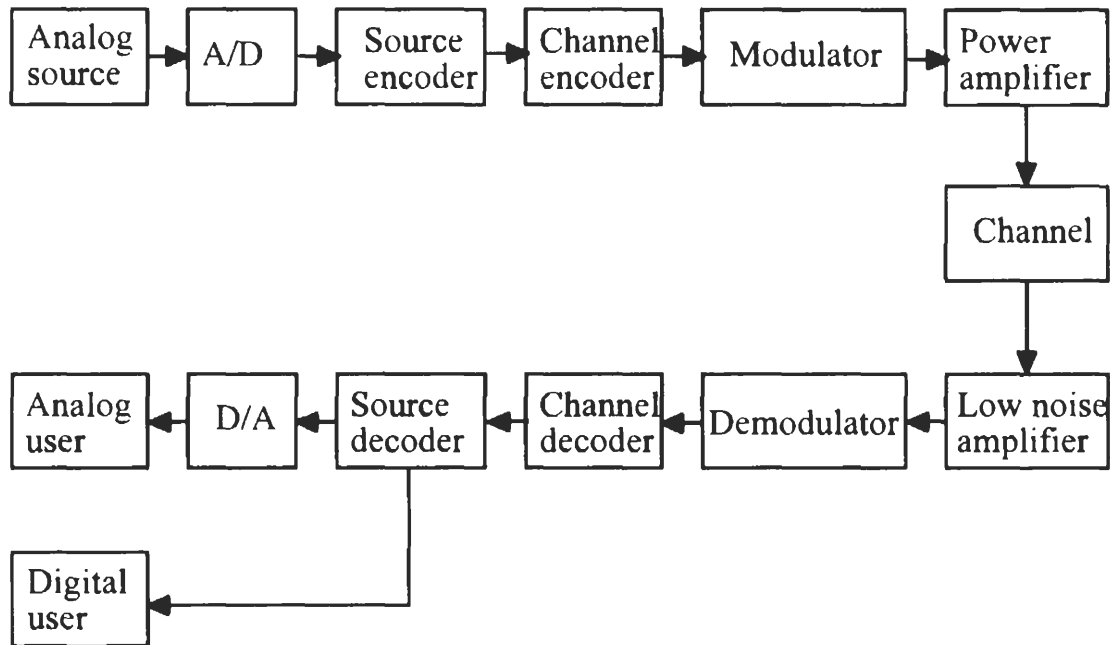


Figure 1.1 Block diagram of a typical digital communication system.

mitter. The transmission medium is usually called the channel, where noise adds to the signal and fading and attenuation effects appear as a complex multiplicative factor on the signal. The term noise here is a wide-sense term which includes all kinds of random electrical disturbance from outside or from within the system. The channel also usually has a limited frequency bandwidth so that it can be viewed as a filter. In the receiver, virtually the reverse signal processing happens. First the received weak signal is amplified (and down-converted if needed) and demodulated. Then the added redundancy is taken away by the channel decoder and the source decoder recovers the signal to its original form before being sent to the user. A digital-to-analog (D/A) converter is needed for analog signals.

The block diagram in Figure 1.1 is just a typical system configuration. A real system configuration could be more complicated. For a multiuser system, a multiplexing stage is inserted before modulator. For a multistation system, a multiple access control stage is inserted before the transmitter. Other features like frequency spread and encryption can also be added into the system. A real system could be simpler too. Source coding and channel coding may not be needed in a simple system. In fact, only the modulator, channel, demodulator, and amplifiers are essential in all communication systems (with antennas for wireless systems).

For the purpose of describing modulation and demodulation techniques and an-

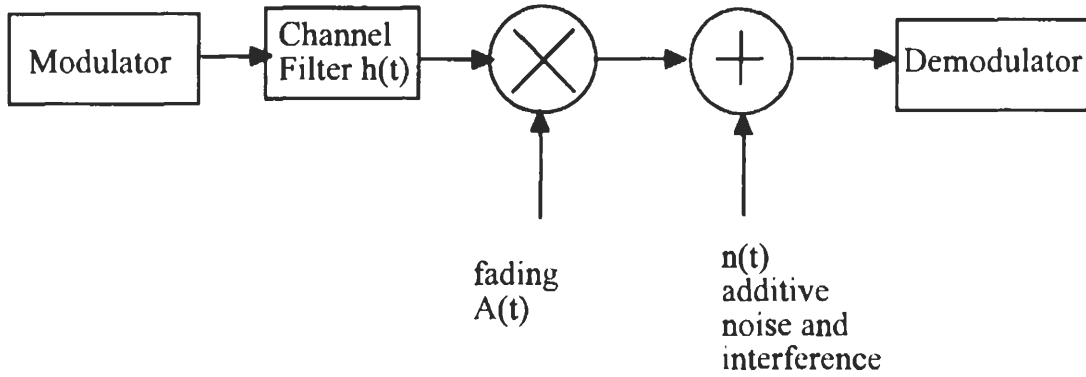


Figure 1.2 Digital communication system model for modulation and demodulation

alyzing their performance, the simplified system model shown in Figure 1.2 will be often used. This model excludes irrelevant blocks with regard to modulation so that relevant blocks stand out. However, recently developed modem techniques combine modulation and channel coding together. In these cases the channel encoder is part of the modulator and the channel decoder is part of the demodulator. From Figure 1.2, the received signal at the input of the demodulator can be expressed as

$$r(t) = A(t) [s(t) * h(t)] + n(t) \quad (1.1)$$

where $*$ denotes convolution. In Figure 1.2 the channel is described by three elements. The first is the channel filter. Because of the fact that the signal $s(t)$ from the modulator must pass the transmitter, the channel (transmission medium) and the receiver before it can reach the demodulator, the channel filter therefore is a composite filter whose transfer function is

$$H(f) = H_T(f)H_C(f)H_R(f) \quad (1.2)$$

where $H_T(f)$, $H_C(f)$, and $H_R(f)$ are the transfer function of the transmitter, the channel, and the receiver, respectively. Equivalently, the impulse response of the channel filter is

$$h(t) = h_T(t) * h_C(t) * h_R(t) \quad (1.3)$$

where $h_T(t)$, $h_C(t)$, and $h_R(t)$ are the impulse responses of the transmitter, the channel, and the receiver, respectively. The second element is the factor $A(t)$ which is generally complex. This factor represents fading in some types of channels, such as mobile radio channel. The third element is the additive noise and interference term $n(t)$. We will discuss fading and noise in more detail in the next section. The channel

model in Figure 1.2 is a general model. It may be simplified in some circumstances, as we will see in the next section.

1.2 COMMUNICATION CHANNELS

Channel characteristic plays an important role in studying, choosing, and designing modulation schemes. Modulation schemes are studied for different channels in order to know their performance in these channels. Modulation schemes are chosen or designed according to channel characteristic in order to optimize their performance. In this section we discuss several important channel models in communications.

1.2.1 Additive White Gaussian Noise Channel

Additive white Gaussian noise (AWGN) channel is a universal channel model for analyzing modulation schemes. In this model, the channel does nothing but add a white Gaussian noise to the signal passing through it. This implies that the channel's amplitude frequency response is flat (thus with unlimited or infinite bandwidth) and phase frequency response is linear for all frequencies so that modulated signals pass through it without any amplitude loss and phase distortion of frequency components. Fading does not exist. The only distortion is introduced by the AWGN. The received signal in (1.1) is simplified to

$$r(t) = s(t) + n(t) \quad (1.4)$$

where $n(t)$ is the additive white Gaussian noise.

The whiteness of $n(t)$ implies that it is a stationary random process with a flat power spectral density (PSD) for all frequencies. It is a convention to assume its PSD as

$$N(f) = N_o/2, \quad -\infty < f < \infty \quad (1.5)$$

This implies that a white process has infinite power. This of course is a mathematical idealization. According to the Wiener-Khinchine theorem, the autocorrelation function of the AWGN is

$$\begin{aligned} R(\tau) &\triangleq E\{n(t)n(t-\tau)\} = \int_{-\infty}^{\infty} N(f)e^{j2\pi f\tau} df \\ &= \int_{-\infty}^{\infty} \frac{N_o}{2} e^{j2\pi f\tau} df = \frac{N_o}{2} \delta(\tau) \end{aligned} \quad (1.6)$$

where $\delta(\tau)$ is the Dirac delta function. This shows the noise samples are uncorrelated

no matter how close they are in time. The samples are also independent since the process is Gaussian.

At any time instance, the amplitude of $n(t)$ obeys a Gaussian probability density function given by

$$p(\eta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{\eta^2}{2\sigma^2}\right\} \quad (1.7)$$

where η is used to represent the values of the random process $n(t)$ and σ^2 is the variance of the random process. It is interesting to note that $\sigma^2 = \infty$ for the AWGN process since σ^2 is the power of the noise, which is infinite due to its “whiteness.”

However, when $r(t)$ is correlated with a orthonormal function $\phi(t)$, the noise in the output has a finite variance. In fact

$$r = \int_{-\infty}^{\infty} r(t)\phi(t)dt = s + n$$

where

$$s = \int_{-\infty}^{\infty} s(t)\phi(t)dt$$

and

$$n = \int_{-\infty}^{\infty} n(t)\phi(t)dt$$

The variance of n is

$$\begin{aligned} E\{n^2\} &= E\left\{\left[\int_{-\infty}^{\infty} n(t)\phi(t)dt\right]^2\right\} \\ &= E\left\{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} n(t)\phi(t)n(\tau)\phi(\tau)dtd\tau\right\} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E\{n(t)n(\tau)\}\phi(t)\phi(\tau)dtd\tau \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{N_o}{2}\delta(t-\tau)\phi(t)\phi(\tau)dtd\tau \\ &= \frac{N_o}{2} \int_{-\infty}^{\infty} \phi^2(t)dt = \frac{N_o}{2} \end{aligned} \quad (1.8)$$

Then the probability density function (PDF) of n can be written as

$$p(n) = \frac{1}{\sqrt{\pi N_o}} \exp\left\{-\frac{n^2}{N_o}\right\} \quad (1.9)$$

This result will be frequently used in this book.

Strictly speaking, the AWGN channel does not exist since no channel can have an infinite bandwidth. However, when the signal bandwidth is smaller than the channel bandwidth, many practical channels are approximately an AWGN channel. For example, the line-of-sight (LOS) radio channels, including fixed terrestrial microwave links and fixed satellite links, are approximately AWGN channels when the weather is good. Wideband coaxial cables are also approximately AWGN channels since there is no other interference except the Gaussian noise.

In this book, all modulation schemes are studied for the AWGN channel. The reason of doing this is two-fold. First, some channels are approximately an AWGN channel, the results can be used directly. Second, additive Gaussian noise is ever present regardless of whether other channel impairments such as limited bandwidth, fading, multipath, and other interferences exist or not. Thus the AWGN channel is the best channel that one can get. The performance of a modulation scheme evaluated in this channel is an upper bound on the performance. When other channel impairments exist, the system performance will degrade. The extent of degradation may vary for different modulation schemes. The performance in AWGN can serve as a standard in evaluating the degradation and also in evaluating effectiveness of impairment-combatting techniques.

1.2.2 Bandlimited Channel

When the channel bandwidth is smaller than the signal bandwidth, the channel is bandlimited. Severe bandwidth limitation causes intersymbol interference (ISI) (i.e., digital pulses will extend beyond their transmission duration (symbol period T_s)) and interfere with the next symbol or even more symbols. The ISI causes an increase in the bit error probability (P_b) or bit error rate (BER), as it is commonly called. When increasing the channel bandwidth is impossible or not cost-efficient, channel equalization techniques are used for combatting ISI. Throughout the years, numerous equalization techniques have been invented and used. New equalization techniques are appearing continuously. We will not cover them in this book. For introductory treatment of equalization techniques, the reader is referred to [1, Chapter 6] or any other communication systems books.

1.2.3 Fading Channel

Fading is a phenomena occurring when the amplitude and phase of a radio signal change rapidly over a short period of time or travel distance. Fading is caused by interference between two or more versions of the transmitted signal which arrive at the receiver at slightly different times. These waves, called multipath waves, combine at the receiver antenna to give a resultant signal which can vary widely in amplitude and phase. If the delays of the multipath signals are longer than a symbol period, these multipath signals must be considered as different signals. In this case, we have individual multipath signals.

In mobile communication channels, such as terrestrial mobile channel and satellite mobile channel, fading and multipath interference are caused by reflections from surrounding buildings and terrains. In addition, the relative motion between the transmitter and receiver results in random frequency modulation in the signal due to different Doppler shifts on each of the multipath components. The motion of surrounding objects, such as vehicles, also induces a time-varying Doppler shift on multipath component. However, if the surrounding objects move at a speed less than the mobile unit, their effect can be ignored [2].

Fading and multipath interference also exist in fixed LOS microwave links [3]. On clear, calm summer evenings, normal atmospheric turbulence is minimal. The troposphere stratifies with inhomogeneous temperature and moisture distributions. Layering of the lower atmosphere creates sharp refractive index gradients which in turn create multiple signal paths with different relative amplitudes and delays.

Fading causes amplitude fluctuations and phase variations in received signals. Multipath causes intersymbol interference. Doppler shift causes carrier frequency drift and signal bandwidth spread. All these lead to performances degradation of modulations. Analysis of modulation performances in fading channels is given in Chapter 10 where characteristics of fading channels will be discussed in more detail.

1.3 BASIC MODULATION METHODS

Digital modulation is a process that impresses a digital symbol onto a signal suitable for transmission. For short distance transmissions, baseband modulation is usually used. Baseband modulation is often called line coding. A sequence of digital symbols are used to create a square pulse waveform with certain features which represent each type of symbol without ambiguity so that they can be recovered upon reception. These features are variations of pulse amplitude, pulse width, and pulse position. Figure 1.3 shows several baseband modulation waveforms. The first one is the non-return to zero-level (NRZ-L) modulation which represents a symbol 1 by a positive

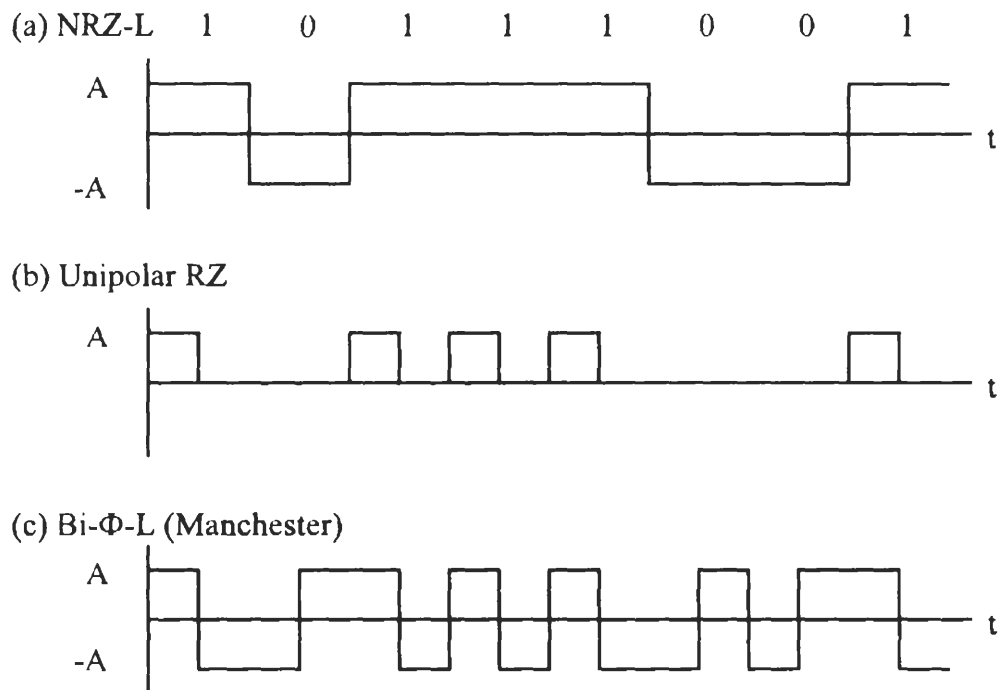


Figure 1.3 Baseband digital modulation examples.

square pulse with length T and a symbol 0 by a negative square pulse with length T . The second one is the unipolar return to zero modulation with a positive pulse of $T/2$ for symbol 1 and nothing for 0. The third is the biphase level or Manchester, after its inventor, modulation which uses a waveform consisting of a positive first-half T pulse and a negative second-half T pulse for 1 and a reversed waveform for 0. These and other baseband schemes will be discussed in detail in Chapter 2.

For long distance and wireless transmissions, bandpass modulation is usually used. Bandpass modulation is also called carrier modulation. A sequence of digital symbols are used to alter the parameters of a high-frequency sinusoidal signal called carrier. It is well known that a sinusoidal signal has three parameters: amplitude, frequency, and phase. Thus amplitude modulation, frequency modulation, and phase modulation are the three basic modulation methods in passband modulation. Figure 1.4 shows three basic binary carrier modulations. They are amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK). In ASK, the modulator puts out a burst of carrier for every symbol 1, and no signal for every symbol 0. This scheme is also called on-off keying (OOK). In a general ASK scheme, the amplitude for symbol 0 is not necessarily 0. In FSK, for symbol 1 a higher frequency burst is transmitted and for symbol 0 a lower frequency burst

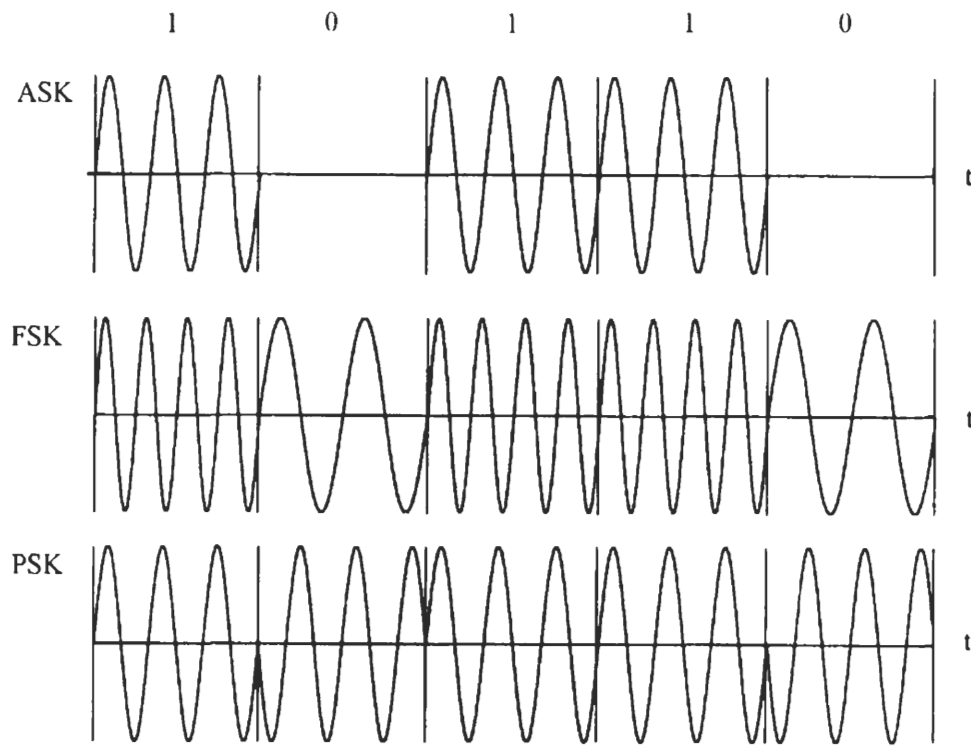


Figure 1.4 Three basic bandpass modulation schemes.

is transmitted, or vice versa. In PSK, a symbol 1 is transmitted as a burst of carrier with 0 initial phase while a symbol 0 is transmitted as a burst of carrier with 180° initial phase.

Based on these three basic schemes, a variety of modulation schemes can be derived from their combinations. For example, by combining two binary PSK (BPSK) signals with orthogonal carriers a new scheme called quadrature phase shift keying (QPSK) can be generated. By modulating both amplitude and phase of the carrier, we can obtain a scheme called quadrature amplitude modulation (QAM), etc.

1.4 CRITERIA OF CHOOSING MODULATION SCHEMES

The essence of digital modem design is to efficiently transmit digital bits and recover them from corruptions from the noise and other channel impairments. There are three primary criteria of choosing modulation schemes: power efficiency, bandwidth

efficiency, and system complexity.

1.4.1 Power Efficiency

The bit error rate, or bit error probability of a modulation scheme is inversely related to E_b/N_o , the bit energy to noise spectral density ratio. For example, P_b of ASK in the AWGN channel is given by

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_o}}\right) \quad (1.10)$$

where E_b is the average bit energy, N_o is the noise power spectral density (PSD), and $Q(x)$ is the Gaussian integral, sometimes referred to as the Q-function. It is defined as

$$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2} du \quad (1.11)$$

which is a monotonically decreasing function of x . Therefore the power efficiency of a modulation scheme is defined straightforwardly as the required E_b/N_o for a certain bit error probability (P_b) over an AWGN channel. $P_b = 10^{-5}$ is usually used as the reference bit error probability.

1.4.2 Bandwidth Efficiency

The determination of bandwidth efficiency is a bit more complex. The bandwidth efficiency is defined as the number of bits per second that can be transmitted in one Hertz of system bandwidth. Obviously it depends on the requirement of system bandwidth for a certain modulated signal. For example, the one-sided power spectral density of an ASK signal modulated by an equiprobable independent random binary sequence is given by

$$\Psi_s(f) = \frac{A^2 T}{4} \text{sinc}^2 [T(f - f_c)] + \frac{A^2}{4} \delta(f - f_c)$$

and is shown in Figure 1.5, where T is the bit duration, A is the carrier amplitude, and f_c is the carrier frequency. From the figure we can see that the signal spectrum stretches from $-\infty$ to ∞ . Thus to perfectly transmit the signal an infinite system bandwidth is required, which is impractical. The practical system bandwidth requirement is finite, which varies depending on different criteria. For example, in Figure 1.5, most of the signal energy concentrates in the band between two nulls, thus a null-to-null bandwidth requirement seems adequate. Three bandwidth efficiencies

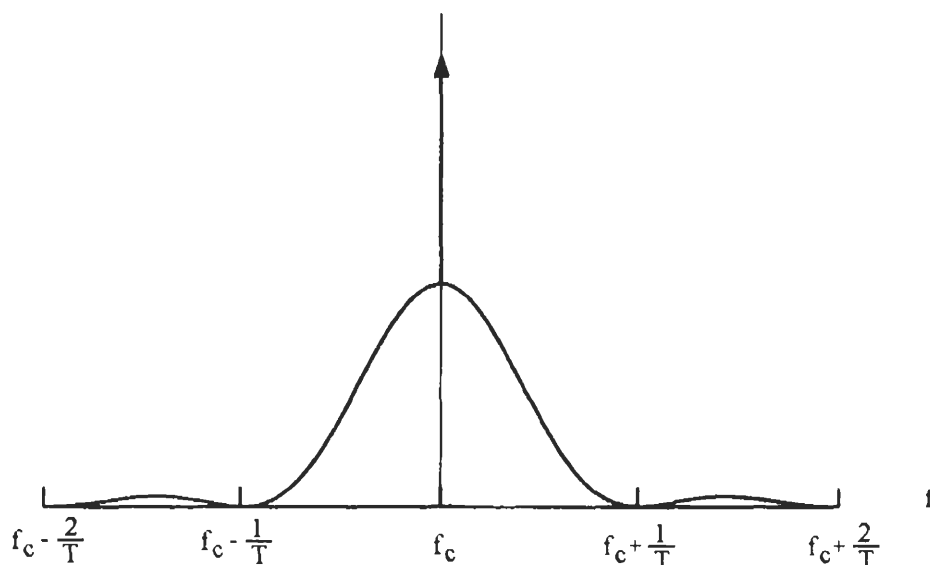


Figure 1.5 Power spectral density of ASK.

used in the literature are as follows:

Nyquist Bandwidth Efficiency—Assuming the system uses Nyquist (ideal rectangular) filtering at baseband, which has the minimum bandwidth required for intersymbol interference-free transmission of digital signals, then the bandwidth at baseband is $0.5R_s$, R_s is the symbol rate, and the bandwidth at carrier frequency is $W = R_s$. Since $R_s = R_b/\log_2 M$, R_b = bit rate, for M-ary modulation, the bandwidth efficiency is

$$R_b/W = \log_2 M \quad (1.12)$$

Null-to-Null Bandwidth Efficiency—For modulation schemes that have power density spectral nulls such as the one of ASK in Figure 1.5, defining the bandwidth as the width of the main spectral lobe is a convenient way of bandwidth definition.

Percentage Bandwidth Efficiency—If the spectrum of the modulated signal does not have nulls, as in general continuous phase modulation (CPM), null-to-null bandwidth no longer exists. In this case, energy percentage bandwidth may be used. Usually 99% is used, even though other percentages (e.g., 90%, 95%) are also used.

1.4.3 System Complexity

System complexity refers to the amount of circuits involved and the technical difficulty of the system. Associated with the system complexity is the cost of manu-

facturing, which is of course a major concern in choosing a modulation technique. Usually the demodulator is more complex than the modulator. Coherent demodulator is much more complex than noncoherent demodulator since carrier recovery is required. For some demodulation methods, sophisticated algorithms like the Viterbi algorithm is required. All these are basis for complexity comparison.

Since power efficiency, bandwidth efficiency, and system complexity are the main criteria of choosing a modulation technique, we will always pay attention to them in the analysis of modulation techniques in the rest of the book.

1.5 OVERVIEW OF DIGITAL MODULATION SCHEMES

To provide the reader with an overview, we list the abbreviations and descriptive names of various digital modulations that we will cover in Table 1.1 and arrange them in a relationship tree diagram in Figure 1.6. Some of the schemes can be derived from more than one “parent” scheme. The schemes where differential encoding can be used are labeled by letter D and those that can be noncoherently demodulated are labeled with a letter N. All schemes can be coherently demodulated.

The modulation schemes listed in the table and the tree are classified into two large categories: constant envelope and nonconstant envelope. Under constant envelope class, there are three subclasses: FSK, PSK, and CPM. Under nonconstant envelope class, there are three subclasses: ASK, QAM, and other nonconstant envelope modulations.

Among the listed schemes, ASK, PSK, and FSK are basic modulations, and MSK, GMSK, CPM, MHPM, and QAM, etc. are advanced schemes. The advanced schemes are variations and combinations of the basic schemes.

The constant envelope class is generally suitable for communication systems whose power amplifiers must operate in the nonlinear region of the input-output characteristic in order to achieve maximum amplifier efficiency. An example is the TWTA (traveling wave tube amplifier) in satellite communications. However, the generic FSK schemes in this class are inappropriate for satellite application since they have very low bandwidth efficiency in comparison with PSK schemes. Binary FSK is used in the low-rate control channels of first generation cellular systems, AMPS (advance mobile phone service of US.) and ETACS (European total access communication system). The data rates are 10 Kbps for AMPS and 8 Kbps for ETACS. The PSK schemes, including BPSK, QPSK, OQPSK, and MSK have been used in satellite communication systems.

The $\pi/4$ -QPSK is worth special attention due to its ability to avoid 180° abrupt phase shift and to enable differential demodulation. It has been used in digital mobile cellular systems, such as the United States digital cellular (USDC) system.

Abbreviation	Alternate Abbr.	Descriptive name
Frequency Shift Keying (FSK)		
BFSK	FSK	Binary Frequency Shift Keying
MFSK		M-ary Frequency Shift Keying
Phase Shift Keying (PSK)		
BPSK	PSK	Binary Phase Shift Keying
QPSK	4PSK	Quadrature Phase Shift Keying
OQPSK	SQPSK	Offset QPSK, Staggered QPSK
$\pi/4$ -QPSK		$\pi/4$ Quadrature Phase Shift Keying
MPSK		M-ary Phase Shift Keying
Continuous Phase Modulations (CPM)		
SHPM		Single-h (modulation index) Phase Modulation
MHPM		Multi-h Phase Modulation
LREC		Rectangular Pulse of Length L
CPFSK		Continuous Phase Frequency Shift Keying
MSK	FFSK	Minimum Shift Keying, Fast FSK
SMSK		Serial Minimum Shift Keying
LRC		Raised Cosine Pulse of Length L
LSRC		Spectrally Raised Cosine Pulse of Length L
GMSK		Gaussian Minimum Shift Keying
TFM		Tamed Frequency Modulation
Amplitude and Amplitude/Phase modulations		
ASK		Amplitude Shift Keying (generic name)
OOK	ASK	Binary On-Off Keying
MASK	MAM	M-ary ASK, M-ary Amplitude Modulation
QAM		Quadrature Amplitude Modulation
Nonconstant Envelope Modulations		
QORC		Quadrature Overlapped Raised Cosine Modulation
SQORC		Staggered QORC
QOSRC		Quadrature Overlapped Squared Raised Cosine Modulation
Q^2 PSK		Quadrature Quadrature Phase Shift Keying
IIF-OQPSK		Intersymbol-Interference/Jitter-Free OQPSK
TSI-OQPSK		Two-Symbol-Interval OQPSK
SQAM		Superposed-QAM
XPSK		Crosscorrelated QPSK

Table 1.1 Digital modulation schemes (Abbr.=Abbreviation).

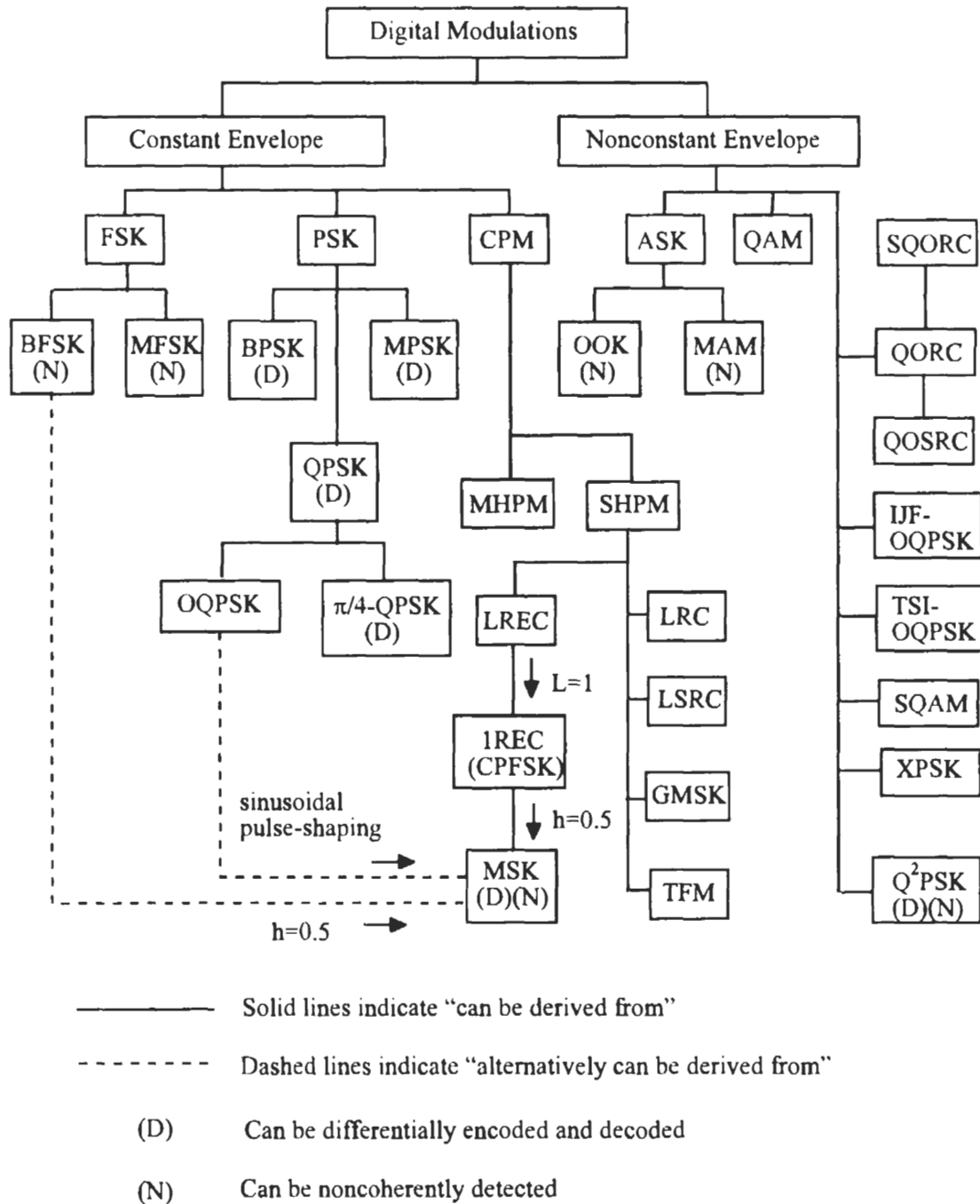


Figure 1.6 Digital Modulation Tree. After [4]

The PSK schemes have constant envelope but discontinuous phase transitions from symbol to symbol. The CPM schemes have not only constant envelope, but also continuous phase transitions. Thus they have less side lobe energy in their spectra in comparison with the PSK schemes. The CPM class includes LREC, LRC, LSRC, GMSK, and TFM. Their differences lie in their different *frequency pulses* which are reflected in their names. For example, LREC means the frequency pulse is a rectangular pulse with a length of L symbol periods. MSK and GMSK are two important schemes in CPM class. MSK is a special case of CPFSK, but it also can be derived from OQPSK with extra sinusoidal pulse-shaping. MSK has excellent power and bandwidth efficiency. Its modulator and demodulator are also not too complex. MSK has been used in NASA's Advanced Communication Technology Satellite (ACTS). GMSK has a Gaussian frequency pulse. Thus it can achieve even better bandwidth efficiency than MSK. GMSK is used in the US cellular digital packet data (CDPD) system and European GSM (global system for mobile communication) system.

MHPM is worth special attention since it has better error performance than single- h CPM by cyclically varying the modulation index h .

The generic nonconstant envelope schemes, such as ASK and QAM, are generally not suitable for systems with nonlinear power amplifiers. However QAM, with a large signal constellation, can achieve extremely high bandwidth efficiency. QAM has been widely used in modems used in telephone networks, such as computer modems. QAM can even be considered for satellite systems. In this case, however, back-off in TWTA's input and output power must be provided to ensure the linearity of the power amplifier.

The third class under nonconstant envelope modulation includes quite a few schemes. These are primarily designed for satellite applications since they have very good bandwidth efficiency and the amplitude variation is minimal. All of them except Q^2 PSK are based on $2T_s$ amplitude pulse shaping and their modulator structures are similar to that of OQPSK. The scheme Q^2 PSK is based on four orthogonal carriers.

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